

Sunny Bagal

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Profile

7th-semester B.Tech Computer Engineering student and analytical developer with hands-on experience building full-stack web applications, decentralized streaming systems, Web3 platforms, and ML-assisted image-processing pipelines. Skilled in JavaScript frameworks, Solana integrations, and scalable backends. Managed end-to-end projects including architecture, API design, and deployment. Strong communicator with a focus on problem-solving and delivering user-focused solutions.

Skills

Programming & Languages: Python, JavaScript, C++, TypeScript, Java, HTML/CSS

Web & Frameworks: Next.js, Node.js, React, Tailwind CSS

Blockchain / Web3: Solana APIs, token-gated UX

Databases & Storage: PostgreSQL, MySQL, MongoDB, Convex

Data & ML: NumPy/Pandas, CNNs (image tasks), model prototyping (Google Colab)

DevOps & Tools: Docker, HAProxy, Apache JMeter (performance testing), VS Code, Android Studio, Git

Other: REST APIs, Prisma ORM, performance/load testing, Android app development

Projects

AI-Driven Defense for Federated Learning - Python, PyTorch, Docker Compose

06/2025 – 10/2025

- Built a cross-silo federated learning framework with label-flipping and backdoor attack modules
- Designed an AI defense engine detecting 66% of malicious clients and reducing attack success rate by 88% via gradient-pattern analysis and anomaly scoring
- Implemented reproducible multi-client setups using Docker Compose for benchmarking

A Lightweight CNN for Real-Time Image Classification - PyTorch, Torchvision

07/2025 – 10/2025

- Engineered a compact CNN using depthwise-separable convolutions and SE feature refinement
- Achieved 80%+ classification accuracy on Animals-10 dataset with reduced model size and inference latency vs. baseline CNNs
- Built full training pipeline with Mixup, OneCycleLR, label smoothing, and early stopping for stable convergence

Blockchain-Based Interactive Music Platform - Next.js, Solana APIs, Prisma, PostgreSQL

04/2025 – 06/2025

- Implemented token-gated live voting for music tracks using Solana API integrations
- Designed real-time backend with Prisma + PostgreSQL enabling millisecond-level queue updates and zero data inconsistency
- Delivered smooth token-holder UX with instant verification (<200ms) and responsive UI rendering

Decentralized Video Streaming Prototype - React, Node.js, Convex, HAProxy, JMeter

02/2025 – 05/2025

- Architected horizontally scalable streaming system using HAProxy load balancing and Convex for distributed state
- Stress-tested with Apache JMeter, achieving 100,000+ concurrent user requests with stable latency under peak load
- Tuned backend resource allocation to minimize bottlenecks and improve concurrency tolerance

DAI-Based Image Compression (CNN + Q-Shift DTCWT) - Python, Colab, DTCWT

01/2025 – 06/2025

- Built CNN encoder-decoder with Q-shift DTCWT achieving 1.0 SSIM (lossless reconstruction) across 7,000 images
- Achieved visually identical reconstruction quality with compact compression size
- Demonstrated superior perceptual metrics and consistency relative to baseline compression models

Leadership & Community Service

Co-Head - Creative Department, NMIMS (Technology & Management) — 1 year

Managed department's social media, boosting engagement by 550%. Produced annual Technology branch articles. Designed official merchandise and led decoration planning for the college's flagship event.

Community Service Volunteer – 20 days (Navi Mumbai)

Coordinated event logistics, on-site operations, and activity execution for assigned initiatives. Managed the organization's social media, including content, posting cadence, and community engagement.

Education

B.Tech, Computer Engineering, Narsee Monjee Institute of Management Studies
CGPA: **2.82 / 4.0** (as of Semester 6)

07/2022 – Present | Navi Mumbai

Certificates

Google GEN AI Study Jam

Cyber Security: From Beginner to Advance

Learn Network Hacking From Scratch (WIFI & Wired)